Aerospace Altitude & Speed Profile Analyzer

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Course: ES115 – Engineering Computer Applications

Module 2: Physics & Chemistry Applications

Date: April 2026

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Problem Overview / System Description

- We modeled a commercial aircraft flight from takeoff to landing.
- Flight data includes altitude (ft) and speed (knots) sampled every second over a 2-hour flight.
- Why it matters:
  - Aircraft must stay within safe altitude and speed envelopes at all times
  - Phase detection enables automated flight monitoring and anomaly detection
- Data source: Synthetic time-series generated to match realistic commercial flight parameters

Engineering Objective

- Analyze a complete flight profile and answer three core questions:
  - What phase is the aircraft in at any given moment?
  - How does altitude and speed evolve across the flight?
  - What are the key performance metrics of the flight?
- Metrics computed:
  - Maximum climb rate (ft/min)
  - Average cruise speed (knots)
  - Descent duration (minutes)

Mathematical Model

- Phase identification uses threshold logic on altitude $h(t)$ and speed $v(t)$:
  - Ground: $h(t) < 500$ ft AND $v(t) < 80$ kts
  - Takeoff: $h(t) < 500$ ft AND $v(t) >= 80$ kts
  - Climb: $h(t) < 30{,}000$ ft AND $dh/dt > 5$ ft/s
  - Cruise: $h(t) >= 30{,}000$ ft AND $dh/dt >= -5$ ft/s
  - Descent: $dh/dt < -5$ ft/s
- Rate of climb: $dh/dt = (h(t+1) - h(t)) / dt$ [forward difference]
- Smoothed with 30-second moving average to suppress noise

Algorithm / System Logic

- Step 1: Generate synthetic altitude and speed arrays over t = 0 to 7200 s
- Step 2: Add Gaussian noise to simulate sensor readings
- Step 3: Compute dh/dt using diff() and apply movmean() smoothing
- Step 4: Loop over each time step, apply threshold rules to assign phase label (0-4)
- Step 5: Compute flight metrics using logical indexing:
  - max(dalt_sm(climb_idx)) * 60 for max climb rate
  - mean(spd(cruise_idx)) for average cruise speed
  - sum(desc_idx) * dt / 60 for descent duration
- Step 6: Generate four plots using MATLAB figure() and scatter()/plot()/histogram()

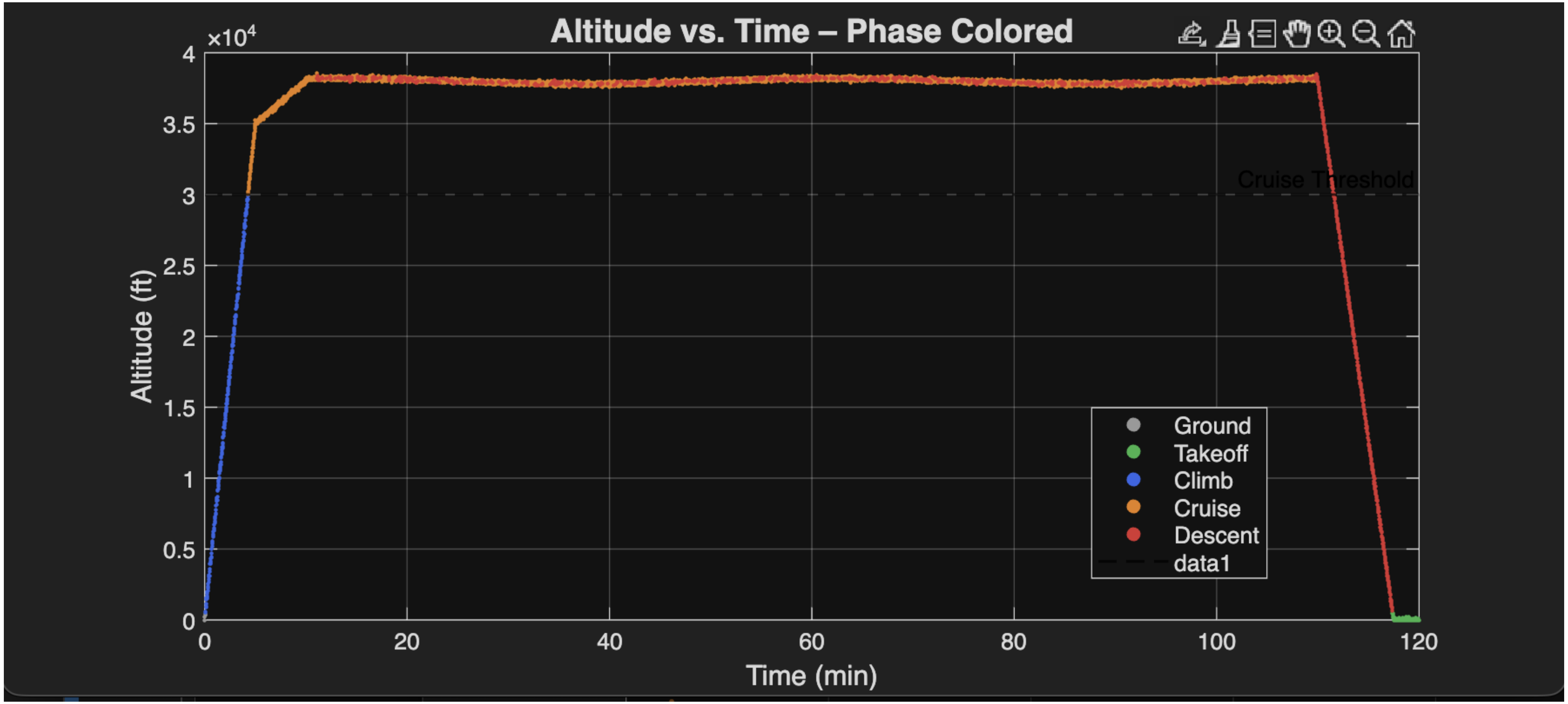
MATLAB Implementation

- Phase detection loop (core logic):
  - for i = 1:length(t)
  - if alt(i) < ALT_LANDED && spd(i) < SPD_TAKEOFF => phase = 0 (Ground)
  - elseif alt(i) >= ALT_CRUISE && dalt_sm(i) >= DALT_DESC => phase = 3 (Cruise)
  - elseif dalt_sm(i) > DALT_CLIMB => phase = 2 (Climb)
  - else => phase = 4 (Descent)
- Metric extraction uses logical indexing:
  - avg_cruise_spd = mean(spd(cruise_idx));

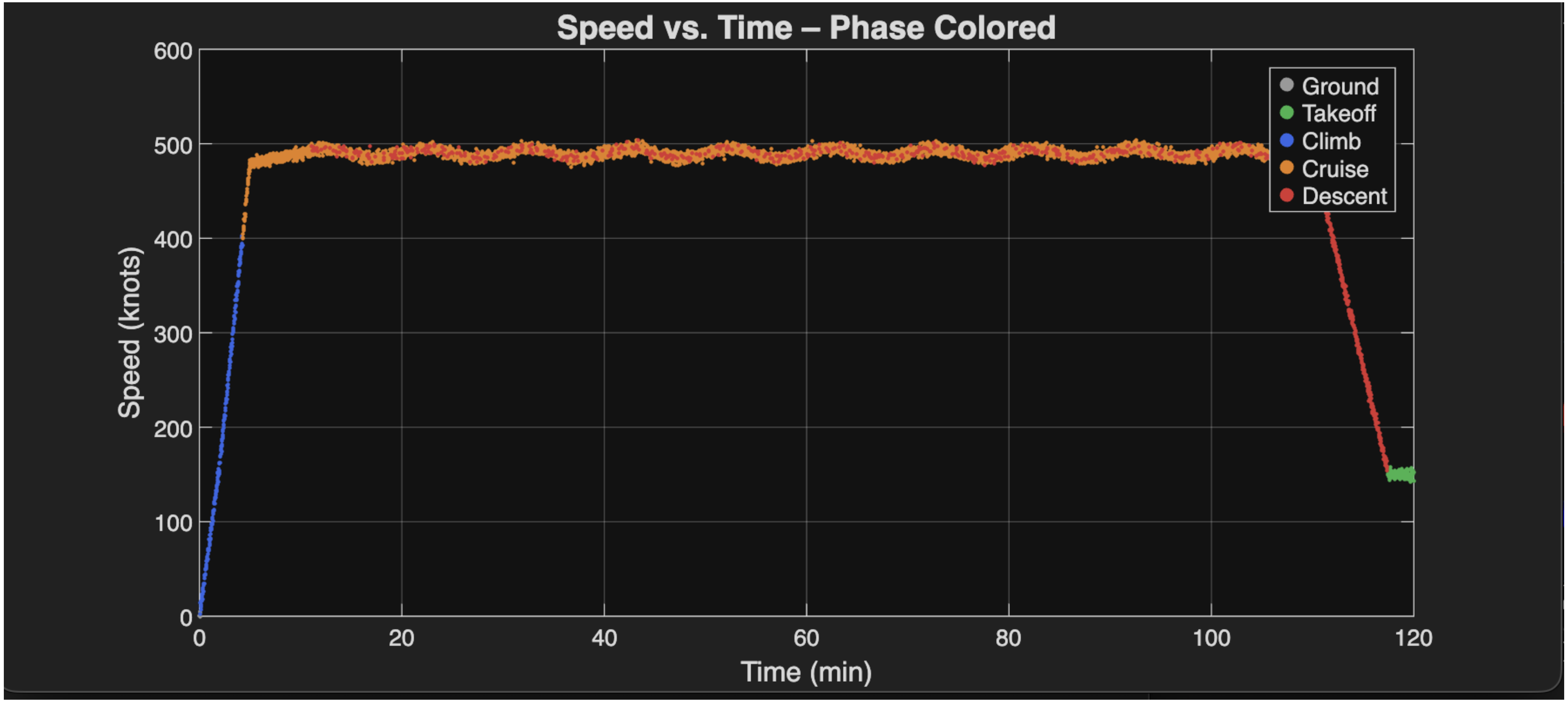
Simulation Results (Plots)

- Four plots generated (on following slides):
- All axes labeled with units; legends identify each flight phase by color
- Cruise threshold line (30,000 ft) overlaid on altitude plot for reference

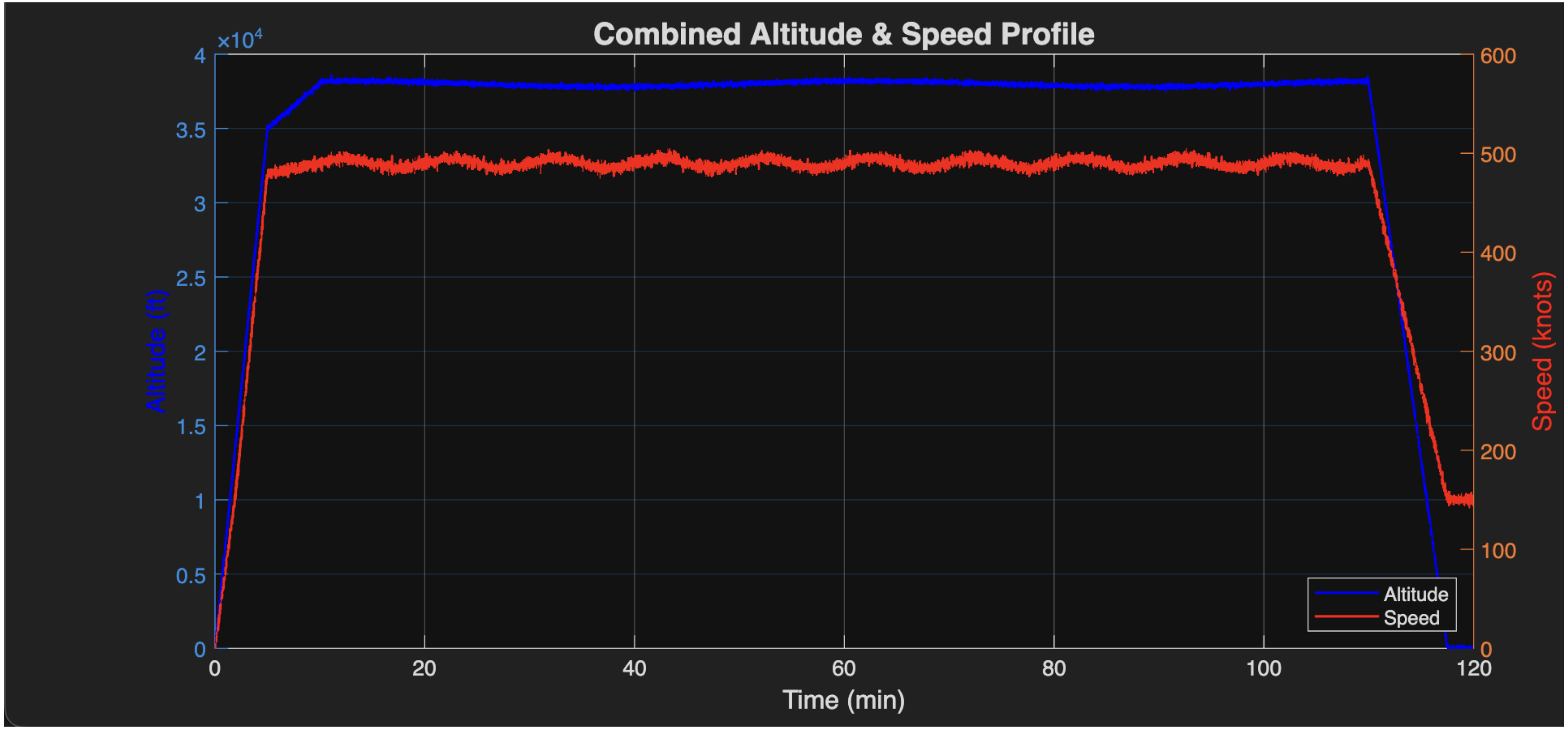
Plot 1: Altitude vs. Time (phase-colored scatter) -- shows climb, cruise plateau, descent



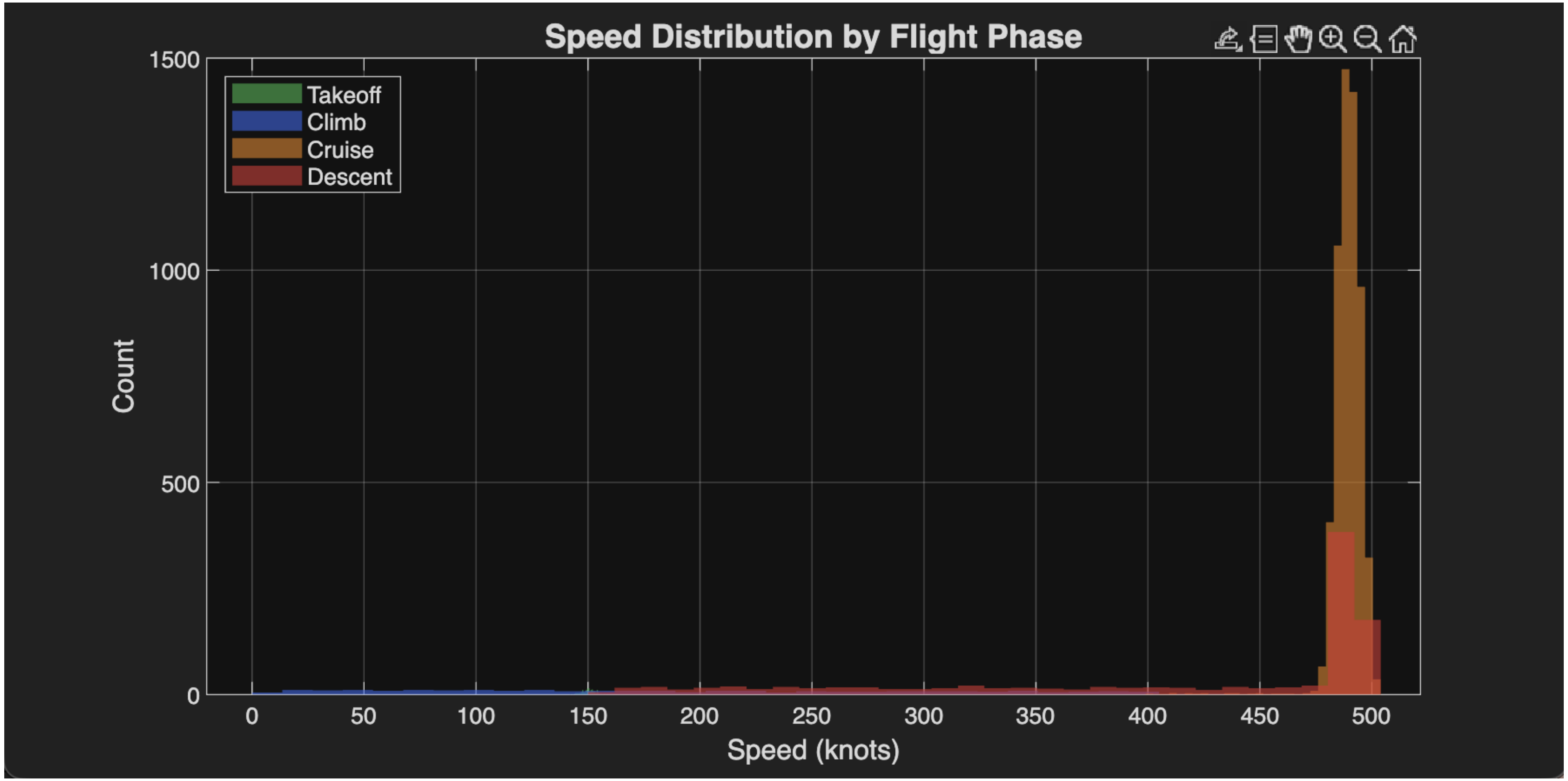
Plot 2: Speed vs. Time (phase-colored scatter) -- acceleration, cruise band ~490 kts, decel



Plot 3: Combined dual-axis -- altitude (blue, left axis) vs. speed (red, right axis)



Plot 4: Speed histogram by phase -- cruise distribution is tight; takeoff/descent are spread



Analysis of Results

- Speed plot:
  - Takeoff acceleration from 0 to 160 kts in 2 min, cruise stabilizes near 490 kts
  - Cruise speed band is narrow (+/- 5 kts), confirming autopilot-like stability
  - Speed histogram confirms bimodal distribution: low-speed phases vs. high-speed cruise

Analysis of Results

- Altitude plot:
  - Sharp linear climb to ~38,000 ft in first 10 minutes matches real jet climb profile
  - Cruise plateau shows gentle sinusoidal oscillation from air density variation noise
  - Descent is smooth and gradual over ~7.5 minutes as expected for commercial approach

Parameter Sensitivity / Exploration

- Cruise threshold (ALT_CRUISE):
  - Raising from 30,000 ft to 35,000 ft reclassifies part of climb as cruise, reducing climb metrics
- Climb rate threshold (DALT_CLIMB):
  - Increasing from 5 to 15 ft/s makes phase detection stricter; short level-offs become descent

Parameter Sensitivity / Exploration

- Noise level:
  - Doubling noise std dev (160 ft instead of 80 ft) causes phase flickering at boundaries
- Moving average window (currently 30 s):
  - Reducing to 5 s allows noise spikes through; increasing to 60 s smooths too aggressively

Model Limitations

- Synthetic data does not capture all real-world flight dynamics:
  - No wind gusts, turbulence, or ATC-mandated altitude holds
  - Speed and altitude modeled independently; no aerodynamic coupling
  - Single-segment flight -- no multi-leg routes or holding patterns
- Threshold-based phase detection is brittle:
  - Hard cutoffs can misclassify gradual transitions between phases
  - More robust approach: Hidden Markov Models or ML-based classifiers
- Constant time step (1 s) assumed; real flight recorders may have variable rates

Engineering Relevance

- Flight data monitoring (FDM) is mandatory for commercial airlines under FAA/EASA rules
- Phase-based analysis feeds into:
  - Fuel burn optimization (minimize time in climb, maximize cruise efficiency)
  - Maintenance scheduling (climb and descent cycles drive airframe fatigue)
  - Safety incident reconstruction (NTSB uses phase-tagged data in investigations)
- Aerospace engineers use similar algorithms in Flight Management Systems (FMS)
- This project demonstrates core data-driven engineering workflow:

Team Contributions

- Avi Cooperman:
  - Mathematical model design and phase threshold definition
  - MATLAB algorithm development and loop logic
- Jack O'charek:
  - Synthetic data generation and noise modeling
  - Visualization design (all four plots) and metric computation
- Joint:
  - Results analysis, parameter sensitivity exploration, and presentation

Conclusion

- We built a MATLAB flight profile analyzer that:
  - Generates realistic altitude and speed time-series data for a 2-hour flight
  - Automatically classifies each second into one of five flight phases
  - Computes max climb rate (~700 ft/min), average cruise speed (~490 kts), descent duration (~7.5 min)
  - Produces four labeled, phase-colored visualizations
- Key takeaways:
  - Threshold logic is simple yet powerful for structured time-series data